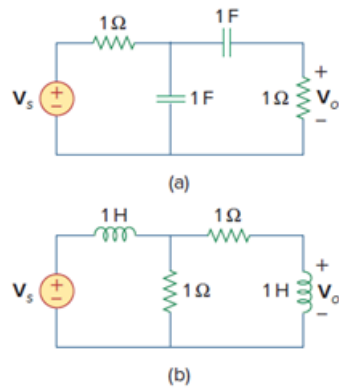


Problem

Determine the center frequency and bandwidth of the band-pass filters in Fig. 14.88.

Figure 14.88:



Step-by-step solution

Step 1 of 8

(a)

Refer to the Figure 14.88 (a) in the textbook.

The modified s-domain circuit is shown in Figure 1.

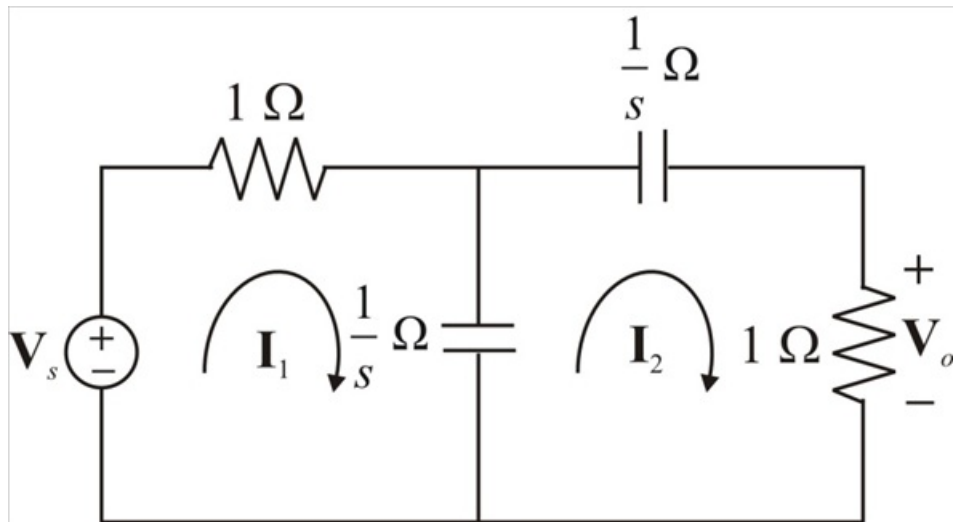


Figure 1

Step 2 of 8

Apply Kirchhoff voltage law to mesh1 in Figure 1.

$$-V_s + 1I_1 + \frac{1}{s}(I_1 - I_2) = 0$$

$$-sV_s + sI_1 + (I_1 - I_2) = 0$$

$$sI_1 + I_1 - I_2 = sV_s$$

$$(1+s)I_1 - I_2 = sV_s \dots\dots (1)$$

Apply Kirchhoff voltage law to mesh2 in Figure 1.

$$\frac{1}{s}(I_2 - I_1) + \frac{1}{s}I_2 + 1I_2 = 0$$

$$I_2 - I_1 + I_2 + sI_2 = 0$$

$$-I_1 + (2+s)I_2 = 0$$

$$I_1 = (2+s)I_2 \dots\dots (2)$$

Step 3 of 8

Substitute equation (2) in equation (1).

$$(1+s)[(2+s)I_2] - I_2 = sV_s$$

$$(s^2 + 3s + 2)I_2 - I_2 = sV_s$$

$$(s^2 + 3s + 2 - 1)I_2 = sV_s$$

$$(s^2 + 3s + 1)I_2 = sV_s$$

$$I_2 = \left(\frac{s}{s^2 + 3s + 1} \right) V_s$$

From Figure 2, output voltage V_o is,

$$V_o = I_2 \times 1$$

$$V_o = I_2$$

Substitute $\left(\frac{s}{s^2 + 3s + 1} \right) V_s$ for I_2 in the above equation.

$$V_o = \left(\frac{s}{s^2 + 3s + 1} \right) V_s$$

$$\frac{V_o}{V_s} = \frac{s}{s^2 + 3s + 1}$$

The transfer function $H(s)$ is,

$$H(s) = \frac{V_o}{V_s} = \frac{s}{s^2 + 3s + 1} \dots\dots (3)$$

The general band-pass filter is,

$$H(s) = \frac{V_o}{V_s} = \frac{sB}{s^2 + sB + \omega_0^2} \dots\dots (4)$$

Step 4 of 8

Compare the denominators of equation (3) with equation (4).

$$\omega_o^2 = 1$$

$$\omega_o = 1 \text{ rad/s}$$

Therefore, the center frequency ω_o is 1rad/s.

$$B = 3$$

Therefore, the band width B is 3rad/s.

Step 5 of 8

(b)

Refer to the Figure 14.88 (b) in the textbook.

The modified s-domain circuit is shown Figure 2.

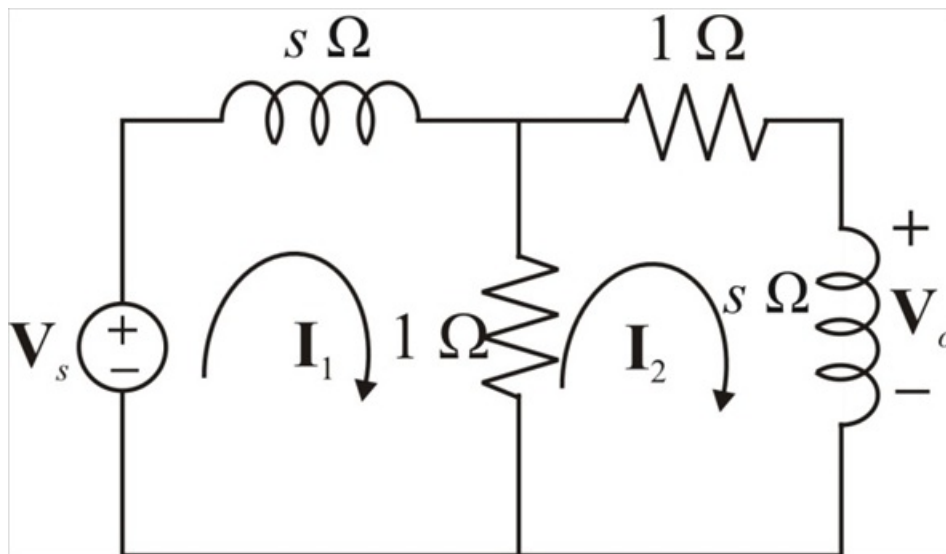


Figure 2

Step 6 of 8

Apply Kirchhoff's voltage law to mesh1 in Figure 2.

$$-V_s + sI_1 + 1(I_1 - I_2) = 0$$

$$(1 + s)I_1 - I_2 = V_s \dots\dots (5)$$

Apply Kirchhoff's voltage law to mesh2 in Figure 2.

$$1(I_2 - I_1) + I_2 + sI_2 = 0$$

$$-I_1 + (2 + s)I_2 = 0$$

$$I_1 = (2 + s)I_2 \dots\dots (6)$$

Step 7 of 8

Substitute equation (6) in equation (5).

$$(1+s)((2+s)\mathbf{I}_2) - \mathbf{I}_2 = \mathbf{V}_s$$

$$(s^2 + 3s + 2)\mathbf{I}_2 - \mathbf{I}_2 = \mathbf{V}_s$$

$$(s^2 + 3s + 2 - 1)\mathbf{I}_2 = \mathbf{V}_s$$

$$(s^2 + 3s + 1)\mathbf{I}_2 = \mathbf{V}_s$$

$$\mathbf{I}_2 = \frac{\mathbf{V}_s}{s^2 + 3s + 1}$$

From Figure 2, the output voltage $\mathbf{V}_o$ is,

$$\mathbf{V}_o = \mathbf{I}_2 \times s$$

$$\mathbf{V}_o = s\mathbf{I}_2$$

Substitute $\frac{\mathbf{V}_s}{s^2 + 3s + 1}$ for $\mathbf{I}_2$ in the above equation.

$$\mathbf{V}_o = s \left(\frac{\mathbf{V}_s}{s^2 + 3s + 1} \right)$$

$$\frac{\mathbf{V}_o}{\mathbf{V}_s} = \frac{s}{s^2 + 3s + 1}$$

The transfer function is,

$$\mathbf{H}(s) = \frac{\mathbf{V}_o}{\mathbf{V}_s} = \frac{s}{s^2 + 3s + 1} \dots\dots (7)$$

The general band-pass filter is

$$\mathbf{H}(s) = \frac{\mathbf{V}_o}{\mathbf{V}_s} = \frac{sB}{s^2 + sB + \omega_0^2} \dots\dots (8)$$

Step 8 of 8

Compare the denominators of equation (7) with equation (8).

$$\omega_o^2 = 1$$

$$\omega_o = 1 \text{ rad/s}$$

Therefore, the center frequency ω_o is 1rad/s.

$$B = 3$$

Therefore, the band width B is 3rad/s.